You-Ting Wu

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Education

Columbia University — New York, NY, USA
Ph.D., Applied Mathematics — Sept 2024 – Aug 2028 (expected)

National Taiwan University — Taipei, Taiwan
B.Sc., Mechanical Engineering and B.Sc., Atmospheric Sciences — Sept 2018 – July 2023

Skills

Machine learning: deep learning, PyTorch, CNN, LSTM, emulation of climate model output, out-of-sample validation
Climate and simulation: climate and fluid dynamics simulation, large-ensemble simulation, nonlinear dynamical systems, numerical PDEs, numerical analysis
Statistics: hypothesis testing, bootstrap resampling, regression, variance decomposition, time series
Programming and HPC: Python (PyTorch, SciPy, Scikit-learn, data visualization), Git, C/C++, MATLAB, Fortran, HPC clusters

Professional Experience

Graduate Research Assistant, Columbia University, *New York, NY, USA* — Sept 2024 – Present

- Built and validated deep learning emulators (CNN and LSTM) in Python (PyTorch) of a climate model, predicting high-dimensional spatiotemporal fields such as cold-spell extremes over 70-year horizons with roughly 100x speedup over direct simulation; evaluated out-of-sample generalization across unseen forcing scenarios to guard against overfitting; manuscript in preparation
- Ran and modified a large Fortran/MPI geophysical fluid dynamics climate model on university HPC clusters, generating a 20-member ensemble of simulations spanning 2,000+ simulated years, and analyzed the output
- Formulated and tested hypotheses on nonlinear, bifurcating dynamics and regime transitions in the climate model under 6 CO2 forcing scenarios, identifying a critical tipping threshold through large-ensemble numerical simulation
- Applied hypothesis testing, regression, and bootstrap resampling to quantify the significance and robustness of simulation results against internal variability (noise)

Software Engineering Intern, Formosa Climate Smart Service, *Taipei, Taiwan* — July 2023 – Sept 2023

- Developed a web platform (Python, HTML/JavaScript) implementing Taiwan's BERSe building energy-efficiency rating: computes building-specific Energy Use Intensity (EUI) scoring scales, actual EUI with an operating-rate correction, and an efficiency grade
- Surveyed carbon footprint disclosures of 11 Taiwanese transport, logistics, and consumer-goods companies, cataloguing reported metrics, life-cycle coverage, and sources to support green finance and sustainability policy work

Visiting Student, Lamont-Doherty Earth Observatory, *Palisades, NY, USA* — June 2022 – Aug 2022

- Applied variance decomposition in Python to roughly 100 ensemble members from 6 climate models (single-model large ensembles), partitioning uncertainty in projected Arctic climate change into internal variability and inter-model spread across seasons
- Applied bootstrap resampling and statistical significance testing to large-ensemble simulations to quantify the robustness of a physical signal against internal variability (noise)

Publications

1. Previdi, M., **Wu, Y.-T.**, Polvani, L.M., England, M.R., Sigmond, M., Liang, Y.-C. Substantial Twentieth-Century Global Sea Level Rise Caused by Ozone-Depleting Substances. Submitted to Journal of Climate (under revision).
2. **Wu, Y.-T.**, Previdi, M., Romanou, A., Lerner, P., Polvani, L.M. Arctic Climate Tipping Points under Delayed CO2 Emissions Mitigation. Communications Earth and Environment.
3. Liang, Y.-C., Polvani, L.M., Previdi, M., Dong, Y., **Wu, Y.-T.**, England, M.R., Griffith, S.M. (2026). The Importance of Non-CO2 Greenhouse Gases to Arctic Warming and Sea-Ice Loss. Geophysical Research Letters, 53, e2026GL122867.
4. **Wu, Y.-T.**, Liang, Y.-C., Previdi, M., Polvani, L.M., England, M.R., Sigmond, M., Lo, M.-H. (2024). Stronger Arctic Amplification from Anthropogenic Aerosols than from Greenhouse Gases. npj Climate and Atmospheric Science, 7(1), 142.
5. **Wu, Y.-T.**, Liang, Y.-C., Kuo, Y.-N., Lehner, F., Polvani, L.M., Previdi, M., Lo, M.-H., Lan, C.-W. (2023). Exploiting SMILEs and the CMIP5 Archive to Understand Arctic Climate Change Seasonality and Uncertainty. Geophysical Research Letters, 50(2), e2022GL100745.

Awards

- Government Scholarship to Study Abroad, Ministry of Education, Taiwan (June 2026)
- Nominated for membership, Sigma Xi, The Scientific Research Honor Society, USA (January 2026)
- Excellent Undergraduate Research Project Award, Ministry of Science and Technology, Taiwan (June 2023)
- Undergraduate Research Project Grant, Ministry of Science and Technology, Taiwan (June 2022)

Professional Service, Presentations, and Teaching

- Journal reviewer: Nature Communications (2026)
- 12 scientific presentations at international conferences (2021–2025)
- 12 teaching assistantships, including Partial Differential Equations, Fluid Dynamics, Numerical Analysis, and Applied Mathematics